IMPLEMENTATION OF DES

AIM:

To write a C program to implement Data Encryption Standard (DES) using C Language.

ALGORITHM:

STEP-1: Read the 64-bit plain text.

STEP-2: Split it into two 32-bit blocks and store it in two different arrays.

STEP-3: Perform XOR operation between these two arrays.

STEP-4: The output obtained is stored as the second 32-bit sequence and the original second 32-bit sequence forms the first part.

STEP-5: Thus the encrypted 64-bit cipher text is obtained in this way. Repeat the same process for the remaining plain text characters.

PROGRAM:

```
import java.util.Scanner; import

java.security.SecureRandom; import

javax.crypto.Cipher; import

javax.crypto.KeyGenerator; import

javax.crypto.SecretKey; import

javax.crypto.spec.SecretKeySpec;


public class Main {


    public static void main(String[] args) throws Exception {


        Scanner sc = new Scanner(System.in);


        System.out.print("Enter Plain Text: ");

        String message = sc.nextLine();
```

```
        KeyGenerator keyGen = KeyGenerator.getInstance("DES");
        SecureRandom random = SecureRandom.getInstance("SHA1PRNG");
random.setSeed("12345".getBytes());

        keyGen.init(56, random);
        SecretKey secretKey = keyGen.generateKey();

        byte[] raw = secretKey.getEncoded();

        System.out.print("DES Key: ");        for (byte
b : raw)
            System.out.printf("%02X", b);
        System.out.println();

        Cipher cipher = Cipher.getInstance("DES");

        cipher.init(Cipher.ENCRYPT_MODE, new SecretKeySpec(raw, "DES"));        byte[]
encrypted = cipher.doFinal(message.getBytes());

        System.out.print("Encrypted Text: ");        for (byte
b : encrypted)
            System.out.printf("%02X", b);
        System.out.println();

        cipher.init(Cipher.DECRYPT_MODE, new SecretKeySpec(raw, "DES"));        byte[]
decrypted = cipher.doFinal(encrypted);
```

```
        System.out.println("Decrypted Text: " + new String(decrypted));        sc.close();
    }
}
```

OUTPUT:

```
Output                                                                Clear
Enter Plain Text: running
DES Key: 01A41F3E49405D7C
Encrypted Text: C4E4BECE1D37D17A
Decrypted Text: running

=== Code Execution Successful ===
```

RESULT:

Thus the data encryption standard algorithm had been implemented successfully using C language.